Academic Year: 2025 – 2026	**Physics Revision Sheet**	**Grade:** 10
Document: Detailed Markscheme	*Light & Kinematics*	**Total:** 20 pts

Exercise 1 : Reflection Angles & Mirror Systems (5 pts)

1. **Combined Angle =** $140°$
   - We know the combined angle is $i + r = 140°$. ***[0.5 pt]***
   - According to the 2nd Law of Reflection, $i = r$. ***[0.5 pt]***
   - Therefore, $2i = 140° \implies i = 70°$ and $r = 70°$. ***[0.5 pt]***
2. **Angle with Surface =** $35°$
   - The normal is perpendicular to the mirror surface ($90°$). ***[0.5 pt]***
   - The angle of reflection is measured from the normal: $r = 90° - 35° = 55°$. ***[0.5 pt]***
   - By the Law of Reflection ($i = r$), the angle of incidence is $i = 55°$. ***[0.5 pt]***
3. **Two-Mirror System (**$M_1 \perp M_2$**)**
   - **At** M_1**:** The ray reflects with $r_1 = i_1 = 50°$. ***[0.5 pt]***
   - **Geometric Deduction:** The reflected ray forms a right-angled triangle with the two mirrors. The angle the ray makes with the surface of M_1 is $90° - 50° = 40°$. ***[0.5 pt]***
   - Since $M_1 \perp M_2$, the angle the ray makes with the surface of M_2 is $90° - 40° = 50°$. ***[0.5 pt]***
   - **At** M_2**:** The angle of incidence i_2 is $90° - 50° = 40°$. Therefore, $r_2 = 40°$. ***[0.5 pt]***

Exercise 2 : Light Beams and Lenses (2 pts)

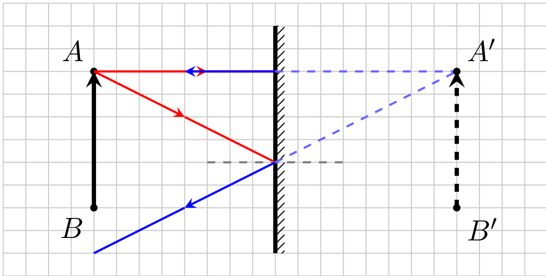
- **Lens A:** Incident beam: **Parallel** ***[0.5 pt]*** ; Emerging beam: **Converging** ***[0.5 pt]***
- **Lens B:** Incident beam: **Parallel** ***[0.5 pt]*** ; Emerging beam: **Diverging** ***[0.5 pt]***

Exercise 3 : Image Formation by a Plane Mirror (5 pts)

1. **Image Types:**
   - Digital camera sensor: **Real image** (it is captured on a physical screen/sensor). ***[0.5 pt]***
   - Bathroom mirror reflection: **Virtual image** (formed behind the mirror by ray extensions). ***[0.5 pt]***
2. **Ray Diagram Construction:**

a) **Point A':** The student must explicitly draw **2 incident rays** originating from A (one normal and one oblique), their corresponding **reflected rays**, and their **virtual extensions** behind the mirror. The virtual extensions (dashed lines) must intersect perfectly at A' (4 units behind the mirror, directly opposite A). ***[1.5 pts]***

b) **Point B':** Plotted symmetrically 4 units behind the mirror, directly opposite B. The virtual object $A'B'$ is drawn as a dashed arrow pointing upwards. ***[1.0 pt]***



c) **Characteristics:** (0.5 pt each) ***[1.5 pts]***

- Virtual (formed behind the mirror).
- Erect / Upright.
- Same size as the object.
- *Also accepted: Symmetric with respect to the mirror.*

Exercise 4 : The Scenic Airplane Trip (4 pts)

- **Phase 1 (Missing Distance):**
  $d_1 = V_1 \times \Delta t_1 = 600$ km/h $\times\ 2$ h $= 1200$ km. ***[0.5 pt]***
- **Phase 2 (Missing Time):**
  $\Delta t_2 = \frac{d_2}{V_2} = \frac{50 \text{ km}}{200 \text{ km/h}} = 0.25$ hours. ***[0.5 pt]***
- **Phase 3 (Missing Time):**
  $\Delta t_3 = \frac{d_3}{V_3} = \frac{600 \text{ km}}{750 \text{ km/h}} = 0.8$ hours. ***[0.5 pt]***
- **Total Distance (d_{tot}):**
  $d_{tot} = d_1 + d_2 + d_3 = 1200 + 50 + 600 =$ **1850** km. ***[0.5 pt]***
- **Total Duration (Δt_{tot}):**
  $\Delta t_{tot} = \Delta t_1 + \Delta t_2 + \Delta t_3 = 2 + 0.25 + 0.8 =$ **3.05** hours. ***[0.5 pt]***
- **Average Speed (V_{avg}):**
  $V_{avg} = \frac{d_{tot}}{\Delta t_{tot}} = \frac{1850}{3.05} \approx$ **606.56** km/h. ***[1.0 pt]***
  Conversion: V_{avg} (m/s) $= \frac{606.56}{3.6} \approx$ **168.49** m/s. ***[0.5 pt]***

Exercise 5 : Motion Recording Analysis (4 pts)

1. **Scenario A:** Uniform rectilinear motion. ***[0.5 pt]***
   Justification: The distance covered by the object during equal and successive time intervals (τ) remains constant (2 cm every step). ***[0.5 pt]***

2. **Scenario B:** Accelerated rectilinear motion. ***[0.5 pt]***
   Justification: The distance covered by the object during equal time intervals (τ) is progressively increasing (e.g., 0.5 cm, then 1.0 cm, then 1.5 cm...). ***[0.5 pt]***

3. **Average Speed (Scenario A):**

   - $v_{avg} = \frac{d}{\Delta t} = \frac{x_5 - x_1}{4\tau}$ ***[0.25 pt]***
   - Position data: $x_1 = 2$ cm, $x_5 = 10$ cm. Duration $= 4 \times 50$ms $= 200$ ms $= 0.2$ s. ***[0.25 pt]***
   - $v_{avg} = \frac{10-2}{0.2} = \frac{8 \text{ cm}}{0.2 \text{ s}} = 40$ cm/s. ***[0.25 pt]***
   - In m/s: $v_{avg} =$ **0.4** m/s. ***[0.25 pt]***

4. **Instantaneous Speed (Scenario B at B_3):**

   - Central difference method: $V_3 = \frac{x_4 - x_2}{2\tau}$ ***[0.25 pt]***
   - Position data (from ruler): $x_2 = 1.5$ cm, $x_4 = 5.0$ cm. Time $= 2 \times 50$ms $= 100$ ms $= 0.1$ s. ***[0.25 pt]***
   - $V_3 = \frac{5.0-1.5}{0.1} = \frac{3.5 \text{ cm}}{0.1 \text{ s}} = 35$ cm/s. ***[0.25 pt]***
   - In m/s: $V_3 =$ **0.35** m/s. ***[0.25 pt]***